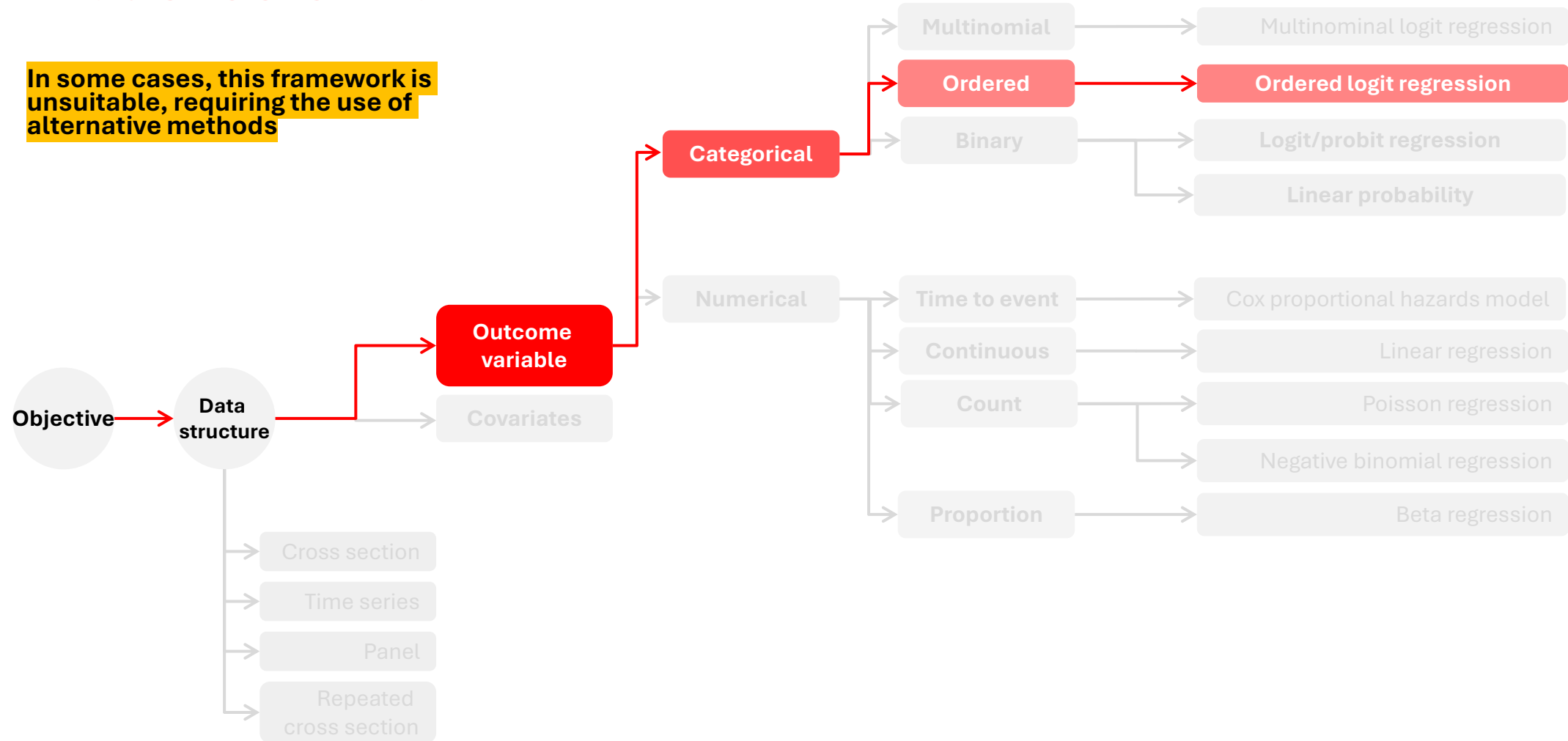


Multinomial models

The ordered logit model

Data types and model selection

In some cases, this framework is unsuitable, requiring the use of alternative methods



Categorical dependent variables with multiple categories

So far, we have studied situations where **the dependent variable is either continuous or binary.**

- **Continuous dependent variables** >
Simple linear regression approaches
- **Binary dependent variables** >
Linear probability models and logistic (logit) models.

In the case of logit models, we looked at scenarios where the dependent variable takes two values:

- Employed ($Y = 1$) or unemployed ($Y = 0$)
- Participation in elections ($Y = 1$) or non-participation ($Y = 0$)

In many cases **the variable of interest can take more than two values:**

1. Nominal categorical dependent variables

$$\text{Region}(x) = \begin{cases} \text{Caribe} & \text{si } x \in \{\text{ATL, BOL, COR, LAG, MAG, SUC, SAP}\} \\ \text{Andina} & \text{si } x \in \{\text{ANT, BOY, CUN, NSA, SAN, TOL, CAL, RIS, QUI, HUT}\} \\ \text{Pacífica} & \text{si } x \in \{\text{CHO, VAC, CAU, NAR}\} \\ \text{Orinoquía} & \text{si } x \in \{\text{ARA, CAS, MET, VIC}\} \\ \text{Amazonía} & \text{si } x \in \{\text{AMA, CAQ, GUV, GUA, PUT, VAU}\} \end{cases}$$

2. Ordinal categorical dependent variables

$$\text{Satisfaction}(x) = \begin{cases} \text{Low} & \text{si } x = 1 \\ \text{Mid} & \text{si } x = 2 \\ \text{High} & \text{si } x = 3 \end{cases} \quad \text{where low} < \text{Mid} < \text{High}$$

Ordinal regression

1. Treat the dependent variable as if it were continuous

In this case, linear regression methods (OLS) can be used.

This approximation depends on the number of categories in the dependent variable.

2. Ignore the ordinality of the dependent variable and treat it as nominal

Here, multinomial logistic models can be used.

In doing so:

- ✗ It discards part of the information available in the ordering
- ✗ It is possible that more parameters than necessary are estimated.
- ✗ It increases the risk of obtaining non-significant statistical results.

3. Treat the dependent variable as ordinal

We assume that the variable is measured on an ordinal scale, even if the scale is a rough measurement of an **underlying interval or ratio scale**.

For example, the categories:

- High
- Medium
- Low

Could be seen as approximations of socioeconomic status or intelligence.

Ordinal regression

Proportional Odds models

1. We observe an ordinal variable Y
2. Y , is a function of another variable, Y^* , **that we can not measure.**

2.1.

We assume that the observed ordinal outcome Y is a function of another variable Y^* , **which is latent, continuous, and unobserved.**

2.2.

In the ordered logistic model, the latent continuous variable Y^* **determines the observed ordinal categories Y .**

2.3.

The latent variable Y^* has M **threshold points** (cut-points), denoted by μ , and the observed value of Y depends on whether these thresholds are crossed.

$$Y_i = \begin{cases} 1 & \text{if } Y_i^* \leq \mu_1 \\ 2 & \text{if } \mu_1 < Y_i^* \leq \mu_2 \\ 3 & \text{if } Y_i^* > \mu_2 \end{cases}$$

Ordinal regression

Y^* : trust in the government

$$Y^* \leq \mu_1 = 5 \quad \rightarrow Y = 1$$

$$(\mu_1 = 5 < Y^* \leq \mu_2 = 15) \quad \rightarrow Y = 2$$

$$(Y^* > \mu_2) \quad \rightarrow Y = 3$$

Ordinal regression

Given the continuous latent variable Y^*

These quantify the effect of each explanatory variable $X_{\{ki\}}$ in the latent variable Y_i^* .

Linear Predictor

This is the systematic part of the model, without the random component.

$$Y_i^* = \alpha + \beta^\top x_k + \varepsilon_i = Z_i + \varepsilon_i$$

Latent Continuous Variable Y^*

- Unobserved for individual i
- It represents the underlying level of a characteristic such as satisfaction, attitude, intention.
- Although we do not observe it directly, we assume that it determines the ordinal category that we can measure.

Explanatory Variables

These are the regressors for individual i .

- Depending on the construct, they may include age, income, or education.
- The subscript k indicates that there are K different variables.

Error Term / Random Disturbance

This captures unobserved factors or measurement errors.

- In ordered logit models, the error term is assumed to follow a standard logistic distribution with mean 0 and variance ≈ 3.29 .
- In ordered probit models, it follows a standard normal distribution, $N(0,1)$.

Ordinal regression

By denoting $\mu = (\mu_0, \mu_1, \mu_2, \dots, \mu_J)$, as a parameter vector with: $\mu_0 = -\infty$ y $\mu_J = +\infty$, the **observation rule for the latent variable is given by:**

$$\begin{cases} y = 1 & \iff \mu_0 \leq \alpha + \beta^T x + \epsilon \leq \mu_1 \\ y = 2 & \iff \mu_1 \leq \alpha + \beta^T x + \epsilon \leq \mu_2 \\ \vdots & \vdots \\ y = J-1 & \iff \mu_{J-2} \leq \alpha + \beta^T x + \epsilon \leq \mu_{J-1} \\ y = J & \iff \mu_{J-1} \leq \alpha + \beta^T x + \epsilon \leq \mu_J \end{cases}$$

Taking $y = 1$, and subtracting α from both sides of the inequality we get:

$$\begin{aligned} y = 1 & \iff \mu_0 \leq \alpha + \beta^T x + \epsilon \leq \mu_1 \\ y = 1 & \iff \mu_0 - \alpha \leq \beta^T x + \epsilon \leq \mu_1 - \alpha \end{aligned}$$

Thus the observation rule does not depend on α , μ_0 , μ_1 or on α , but rather on: $(\mu_0 - \alpha)$ y $(\mu_1 - \alpha)$

$$\begin{aligned} y = 1 & \iff \mu_0 - \alpha - \beta^T x \leq \epsilon \leq \mu_1 - \alpha - \beta^T x \\ y = 1 & \iff \mu_0 - \beta^T x \leq \epsilon \leq \mu_1 - \beta^T x \end{aligned}$$

The decision regarding the value that y takes is therefore reduced to determining where the error term ϵ falls with respect to the thresholds adjusted by the linear predictor.

The model transforms the original thresholds into shifted thresholds, and the probability of each category is calculated as the area under the curve of $\epsilon \sim N(0,1)$ between those limits.

Then we can define F as the cumulative density function of ϵ , for a given j value of y .

$$P(y_n = j) = F(\mu_j - \beta^T x_n) - F(\mu_{j-1} - \beta^T x_n)$$

$$\frac{\partial \Pr[y_n = j]}{\partial x_n} = \{F'(\mu_j - \beta^T x_n) - F'(\mu_{j-1} - \beta^T x_n)\}$$

Ordinal regression

$$P(y_n = j) = F(\mu_j - \beta^\top x_n) - F(\mu_{j-1} - \beta^\top x_n)$$

The probability that observation i falls into the ordered category j is defined as the **difference between the cumulative distribution functions of ε , evaluated at different threshold points μ**

Cumulative distribution functions of ε , evaluated at different threshold points μ , the covariates x , and the parameters β .

The parameters β , along with the $m - 1$ threshold parameters $\mu_1, \dots, \mu_{m-1}$ are obtained by maximising the log-likelihood function of the model.

$$F(z) = \frac{e^z}{1+e^z}$$

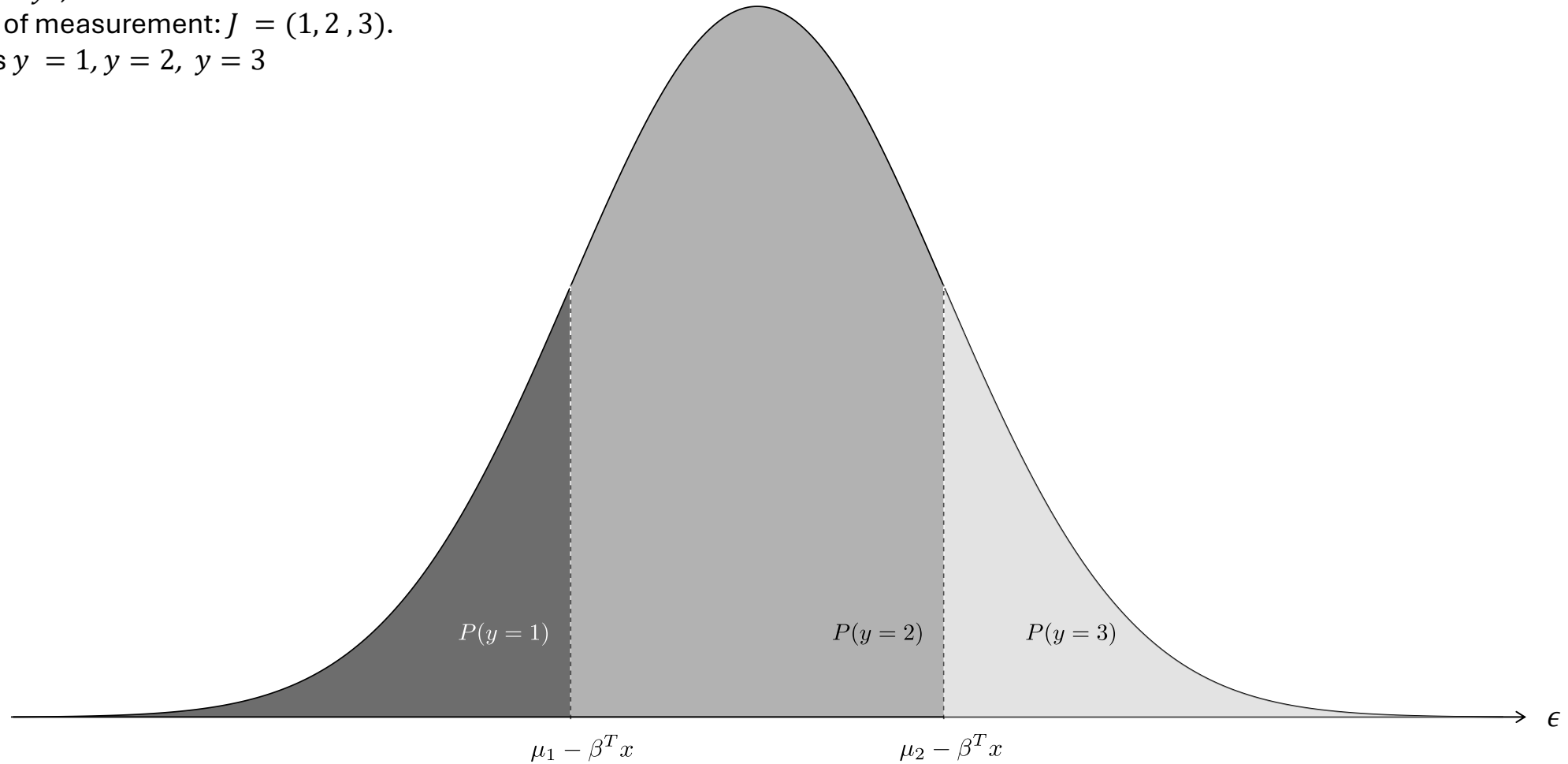
In the ordered logit model, the error term ε follows a **logistic distribution**.

$$F(z) = \Phi(z)$$

In the ordered probit model, the error term ε **follows a standard normal distribution**, and F corresponds to the CDF of that normal distribution

Ordinal regression

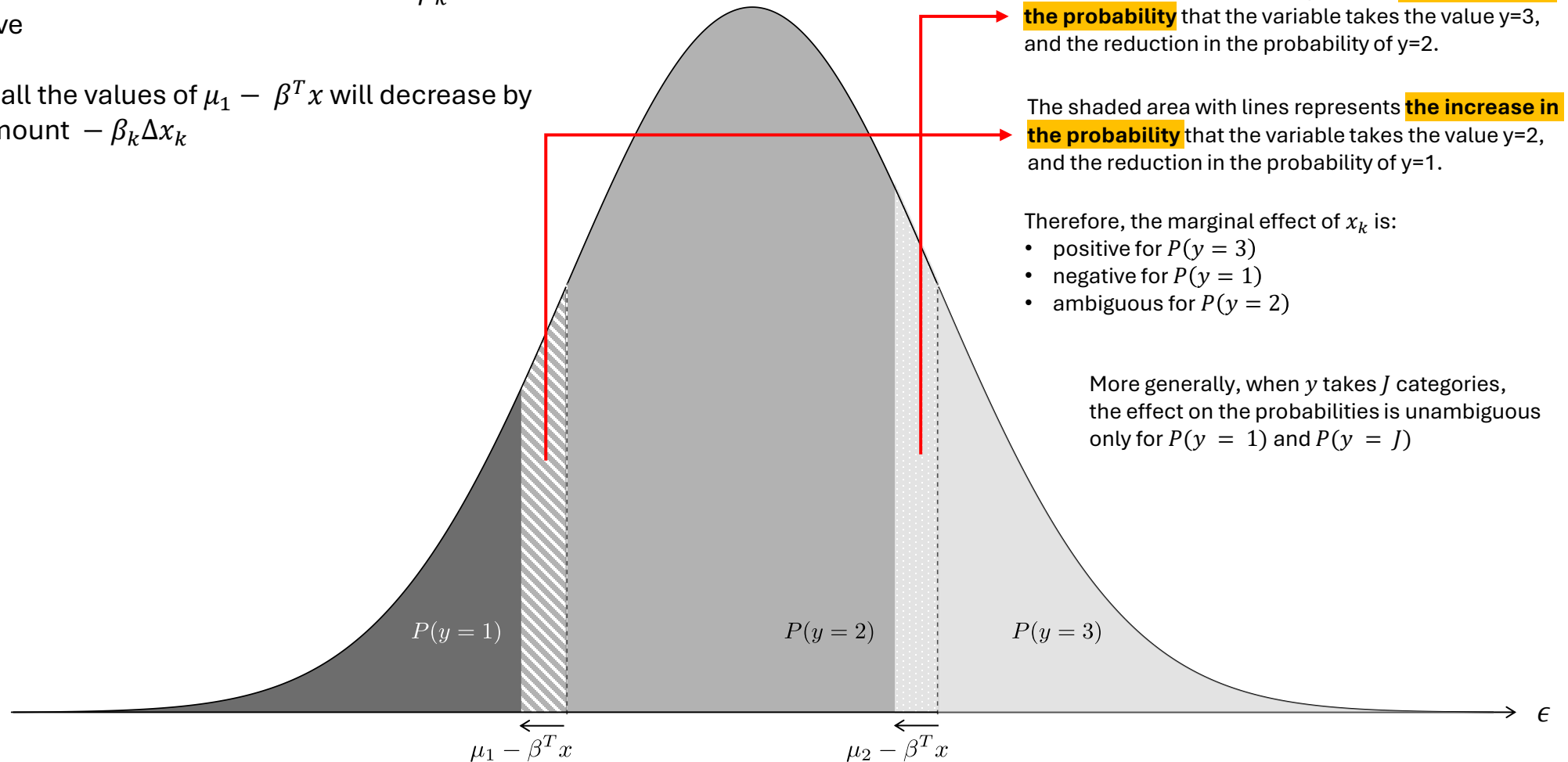
Let us suppose that for a latent variable y^* , we can observe three levels of measurement: $J = (1, 2, 3)$. That is $y = 1, y = 2, y = 3$



Ordinal regression

Consider now an increase in the covariate x_k , and suppose that its associated coefficient β_k is positive

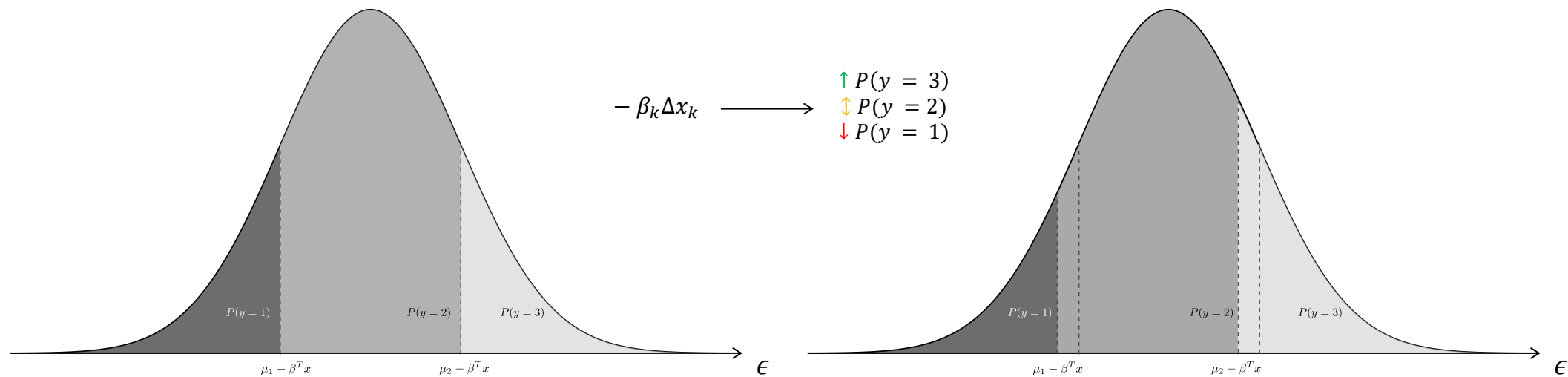
Then, all the values of $\mu_1 - \beta^T x$ will decrease by the amount $-\beta_k \Delta x_k$



Ordinal regression

Consider now an increase in the covariate x_k , and suppose that its associated coefficient β_k is positive

Then, all the values of $\mu_1 - \beta^T x$ will decrease by the amount $-\beta_k \Delta x_k$



Ordinal regression

Logistic link

$$P(Y_n > j) = \frac{e^{\beta^\top x_n - \mu_j}}{1 + e^{(\beta^\top x_n - \mu_j)}}$$

→ The probability that observation i falls into a category higher than j

$$P(Y_n = 1) = 1 - \frac{e^{\beta^\top x_n - \mu_1}}{1 + e^{(\beta^\top x_n - \mu_1)}}$$

→ Probability that observation i falls into category $j = 1$.

$$P(Y_n = j) = \frac{e^{\beta^\top x_n - \mu_{j-1}}}{1 + e^{(\beta^\top x_n - \mu_{j-1})}} - \frac{e^{\beta^\top x_n - \mu_j}}{1 + e^{(\beta^\top x_n - \mu_j)}}$$

→ Probability that observation i falls into category j .

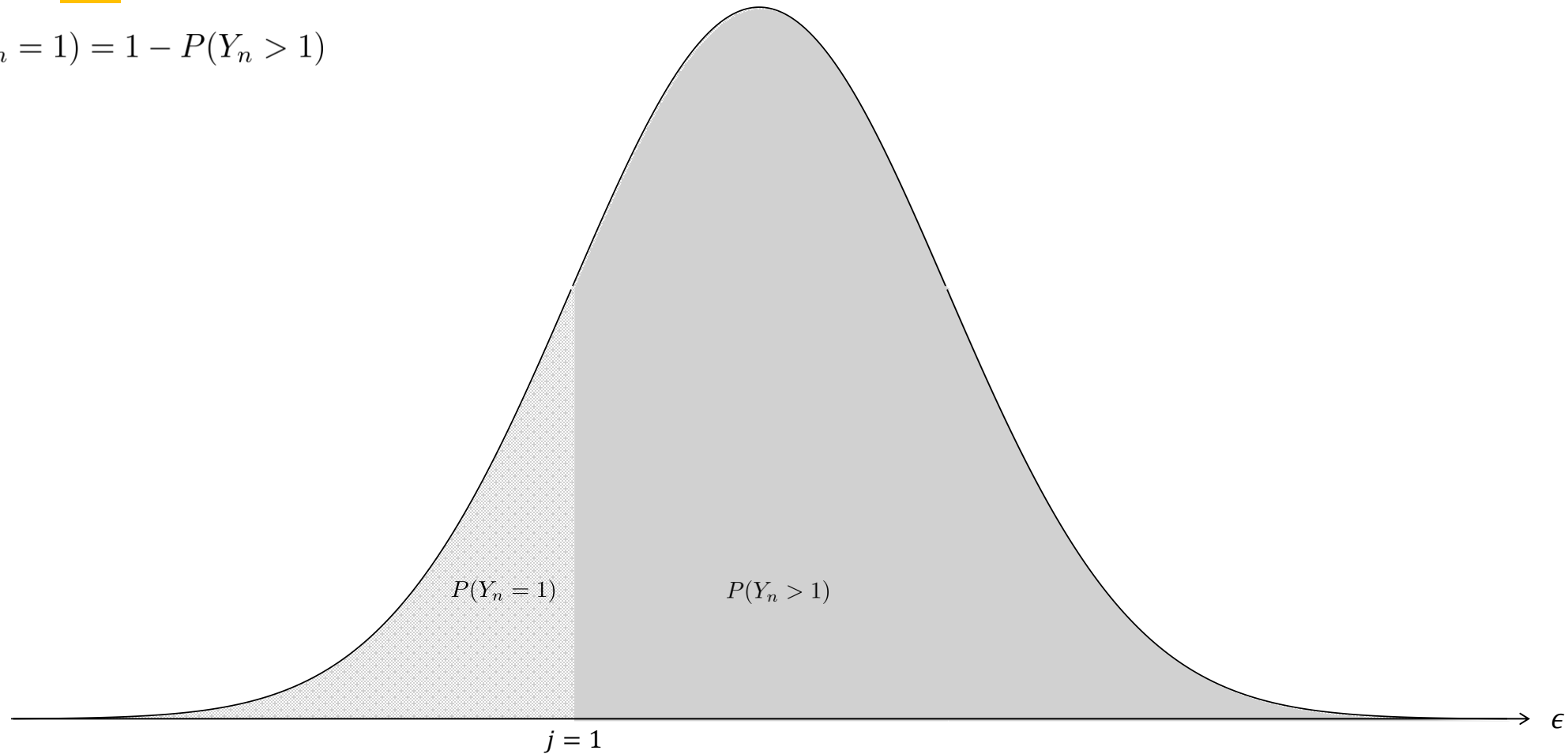
$$P(Y_n = m) = \frac{e^{\beta^\top x_n - \mu_{m-1}}}{1 + e^{(\beta^\top x_n - \mu_{m-1})}}$$

→ Probability that observation i falls into category m .

Ordinal regression

Probability that observation i falls into category **$j = 1$** .

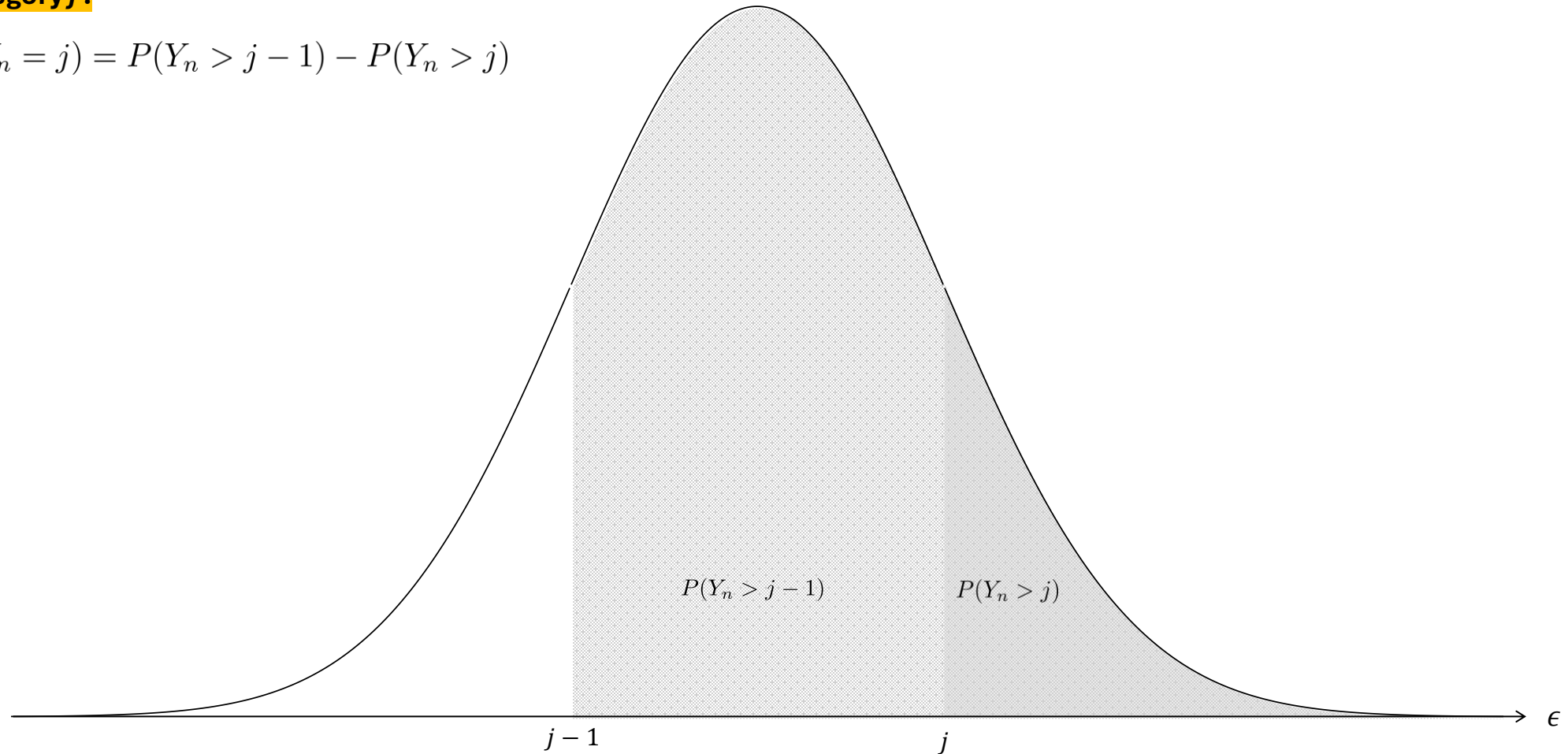
$$P(Y_n = 1) = 1 - P(Y_n > 1)$$



Ordinal regression

Probability that observation i falls into
category j .

$$P(Y_n = j) = P(Y_n > j - 1) - P(Y_n > j)$$



Ordinal regression

The specification of the ordinal logistic model is

$$P(Y_n > j) = \frac{e^{\beta^\top \mathbf{x}_n - \mu_j}}{1 + e^{(\beta^\top \mathbf{x}_n - \mu_j)}}$$

To obtain the odds, we need the probability of the complementary class — that is, the probability that Y_n falls into category j or any lower category ($Y_n \leq j$):

$$P(Y_n \leq j) = 1 - P(Y_n > j)$$

Substituting:

$$P(Y_n \leq j) = 1 - \left(\frac{e^{\beta^\top \mathbf{x}_n - \mu_j}}{1 + e^{\beta^\top \mathbf{x}_n - \mu_j}} \right)$$

$$P(Y_n \leq j) = \frac{1}{1 + e^{\beta^\top \mathbf{x}_n - \mu_j}}$$

The lower cumulative odds ($Y_n \leq j$) are defined as the ratio between:

- The probability that Y_n is greater than j
- The probability that Y_n is less than or equal to j

$$\text{Odds}(Y_n \leq j) = \frac{P(Y_n \leq j)}{P(Y_n > j)}$$

Substituting:

$$\text{Odds}(Y_n \leq j) = \frac{\left(\frac{1}{1 + e^{\beta^\top \mathbf{x}_n - \mu_j}} \right)}{\left(\frac{e^{\beta^\top \mathbf{x}_n - \mu_j}}{1 + e^{\beta^\top \mathbf{x}_n - \mu_j}} \right)}$$

$$\text{Odds}(Y_n \leq j) = \frac{1}{e^{(\beta^\top \mathbf{x}_n - \mu_j)}}$$

By applying the natural logarithm to both sides of the equation, we obtain the log-odds:

$$\ln \left(\frac{P(Y_n \leq j)}{P(Y_n > j)} \right) = \mu_j - \beta^\top \mathbf{x}_n$$

This is the standard form of the **Ordinal Logistic Model with Proportional Odds**, using the lower cumulative logit specification.

Ordinal regression

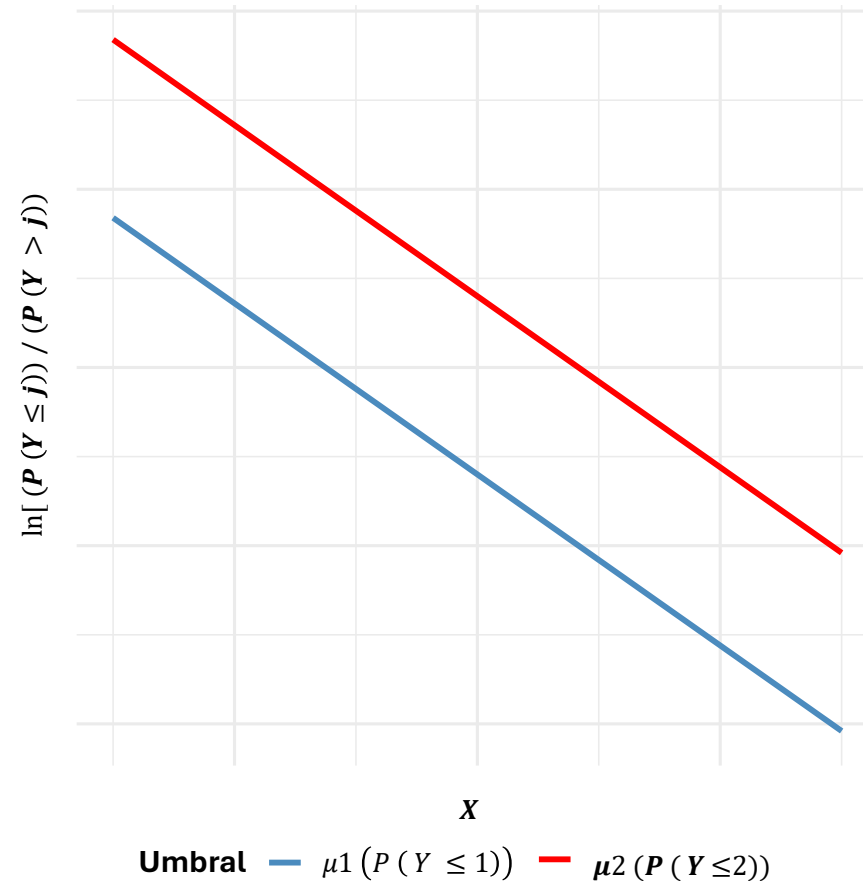
Assumption of Proportional Odds

This assumption states that the effect of any predictor variable (X) on the cumulative odds of obtaining a higher response ($Y > j$) is the same for all threshold points (μ_j) that separate the categories.

$$\ln \left(\frac{P(Y_n \leq j)}{P(Y_n > j)} \right) = \mu_j - \beta^\top \mathbf{x}_n$$

In other words, the coefficient vector β is constant for all levels $j = 1, 2, \dots, M - 1$. The only term that varies across categories is the intercept (cut-point) μ_j .

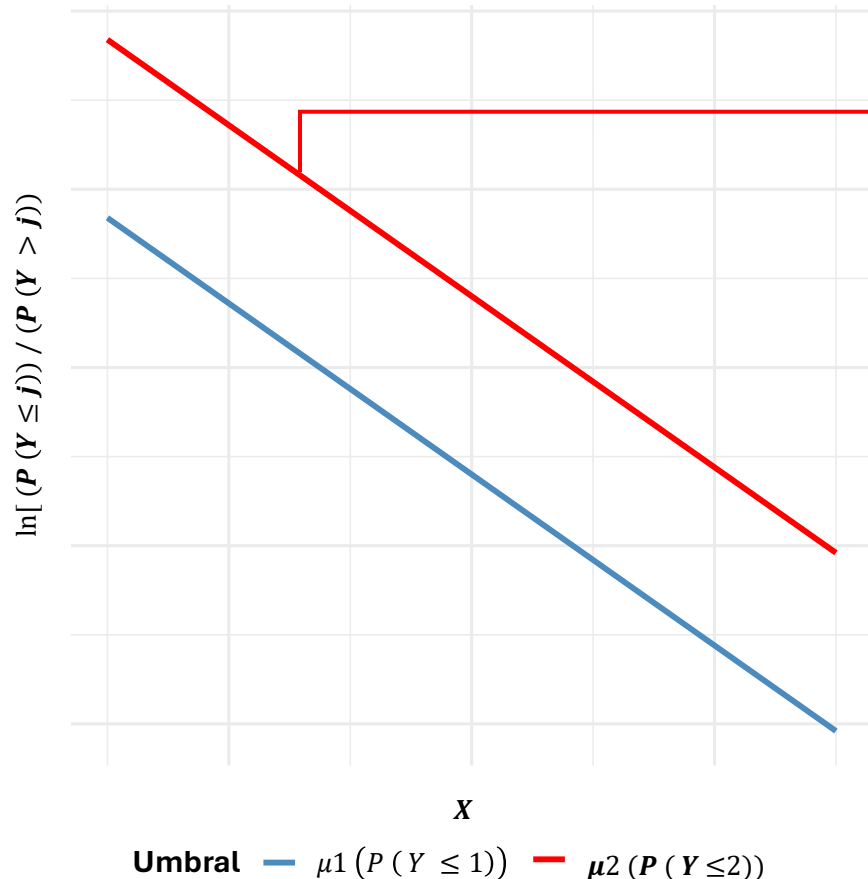
Visualizing the Proportional Odds Assumption



Ordinal regression

Assumption of Proportional Odds

Visualizing the Proportional Odds Assumption



The slope is the same for both thresholds: $-\beta = -1.2$.

Since β is positive (1.2), the logit of $P(Y \leq j)$ is decreasing with respect to X .

An increase in the predictor X (for example, higher GPA) shifts the probabilities towards higher categories, and this happens for both thresholds shown.

As X increases, the odds ratio:

$$\ln \left(\frac{P(Y_n \leq j)}{P(Y_n > j)} \right) = \mu_j - \beta^\top \mathbf{x}_n$$

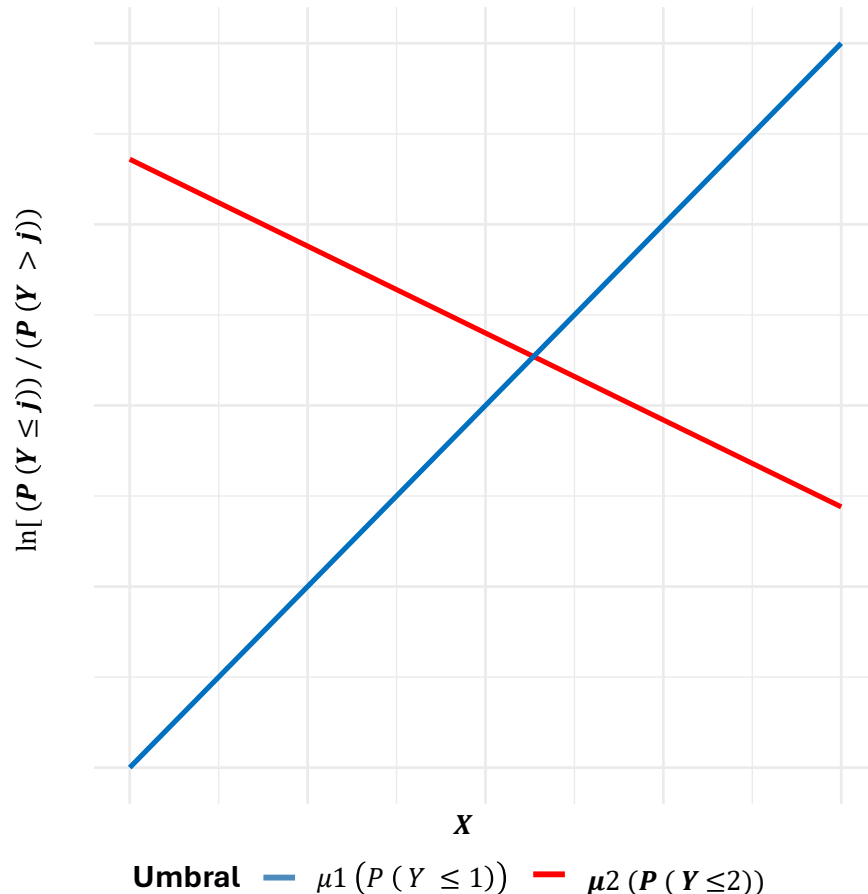
Decreases (becoming smaller than 1), which means that the numerator ($P(Y \leq j)$) falls relative to the denominator ($P(Y > j)$):

This means that an increase in X reduces the odds of obtaining a lower category.

Ordinal regression

Assumption of Proportional Odds

Visualizing the Proportional Odds Assumption



The slope is not the same across thresholds:

- For threshold μ_2 : $-\beta = -1.2$
- For threshold μ_1 : $-\beta = 2.5$
- Since β for threshold 1 is positive (1.2), the logit of $P(Y \leq j)$ is decreasing with respect to X .
- However, in this case, the slope at threshold 2 is positive (+2.5), indicating that the logit of $P(Y \leq j)$ behaves differently across thresholds.

In this situation, the ordered logit model is not suitable for studying the dependent variable, because the model estimates a single coefficient β for all cut-points.

If the true slopes differ, the single estimated slope becomes an average that fits poorly at the extremes of the ordinal scale.

When this happens, one must make decisions about the model's application:

- ✗ Re-categorise the variable, if theoretically justified, and fit a new ordered logit model, or a binary logit based on the new categorisation.
- ✗ If theoretically justified, fit a multinomial logit model.
- ✗ Fit a partial proportional odds logit model, which relaxes the proportional-odds assumption.

Ordinal regression

A researcher is interested in how variables such as grade point average (GPA), the prestige of the undergraduate institution, the type of institution, and parents' education **influence the decision to apply to a postgraduate programme.**

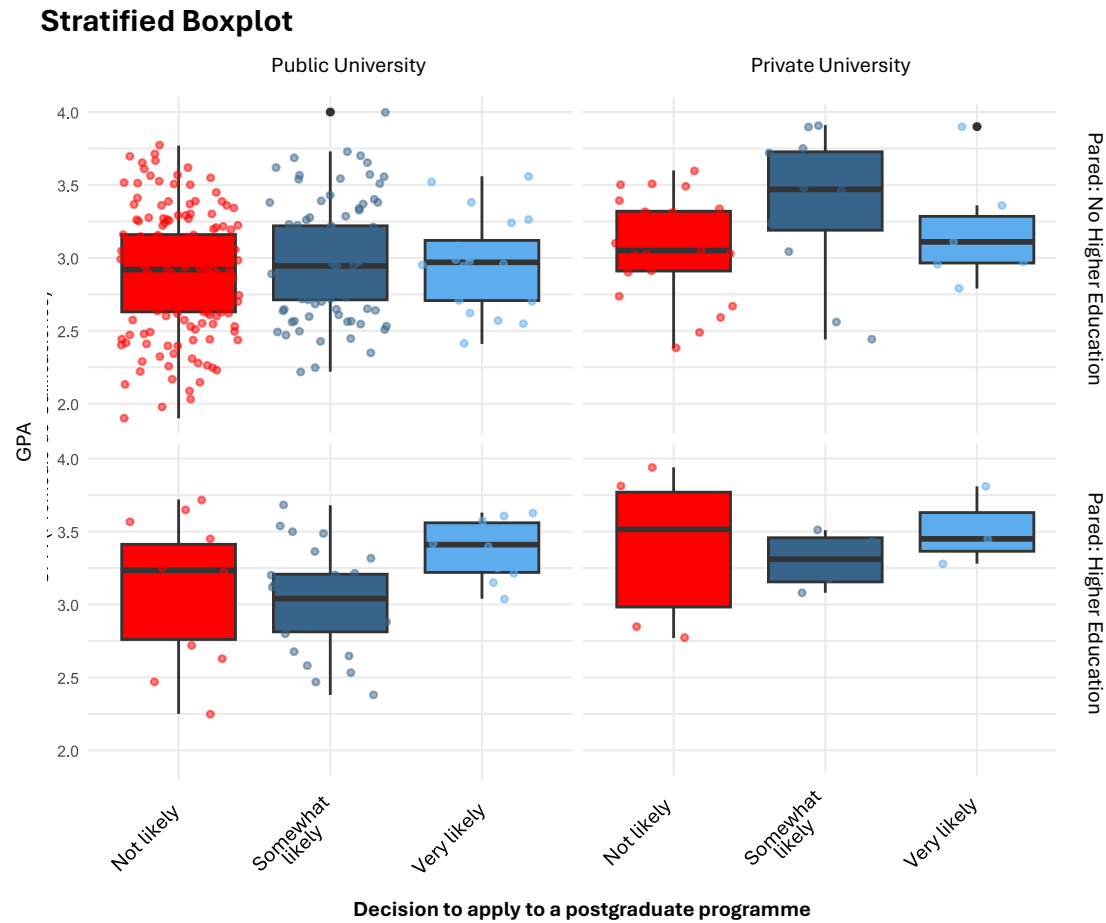
The response variable is **ordinal**:

Apply: {*not likely* < *somewhat likely* < *very likely*}

	apply	pared	public	gpa
1	very likely	0	0	3.26
2	somewhat likely	1	0	3.21
3	unlikely	1	1	3.94
4	somewhat likely	0	0	2.81
5	somewhat likely	0	0	2.53
6	unlikely	0	1	2.59
7	somewhat likely	0	0	2.56

- **Apply.** Decision to apply to a postgraduate programme
- **Pared.** Indicator variable specifying whether the student's parents hold an undergraduate degree.
- **Public.** Indicator variable denoting whether the student completed their undergraduate studies at a public university.
- **GPA.** Independent numerical variable describing the student's cumulative academic average (Grade Point Average).

Ordinal regression



Relative frequencies table

Category	Level	Frequency	R frequency
A. Intention to Apply (apply)			
Not likely	1	220	55%
Somewhat likely	2	140	35%
Very likely	3	40	10%
B. Parents' Higher Education (pared)			
No Higher Education	0	337	84.25%
With Higher Education	1	63	15.75%
C. Type of University (public)			
Private	0	343	85.75%
Public	1	57	14,25%

Ordinal regression

When estimating the model, we obtain:

Regression function used

Dependent variable

Independent variables

Reference to the table containing the variables

Estimated coefficients of the ordinal logit model.

Their scale is in log-odds, which means they cannot be interpreted in the same way as coefficients from an OLS regression. To interpret them correctly, researchers must:

- compute odds ratios, and
- evaluate predicted probabilities.

Null deviance (model without predictors).

Indicates how well (or poorly) the model predicts the response variable when no control variables are included. Higher values imply a poorer fit.

```
Call:
MASS::polr(formula = apply ~ pared + public + gpa, data = dat,
Hess = TRUE)

Coefficients:
              Value Std. Error t value
pared      1.04769    0.2658   3.9418
public    -0.05879    0.2979  -0.1974
gpa        0.61594    0.2606   2.3632

Intercepts:
              Value Std. Error t value
unlikely|somewhat likely  2.2039  0.7795   2.8272
somewhat likely|very likely 4.2994  0.8043   5.3453

Residual Deviance: 717.0249
AIC: 727.0249
```

Test statistic. Evaluates whether each coefficient differs significantly from zero.

To reject the null hypothesis associated with a coefficient, the value of the statistic must exceed the critical value corresponding to the chosen significance level α (1.96 when $\alpha = 0.05$).

Intercepts (cut-points) of the model.

These are not direct probabilities. Instead, they correspond to the threshold parameters (μ_k) on the latent, underlying continuous scale (Y^*), expressed on the logit scale.

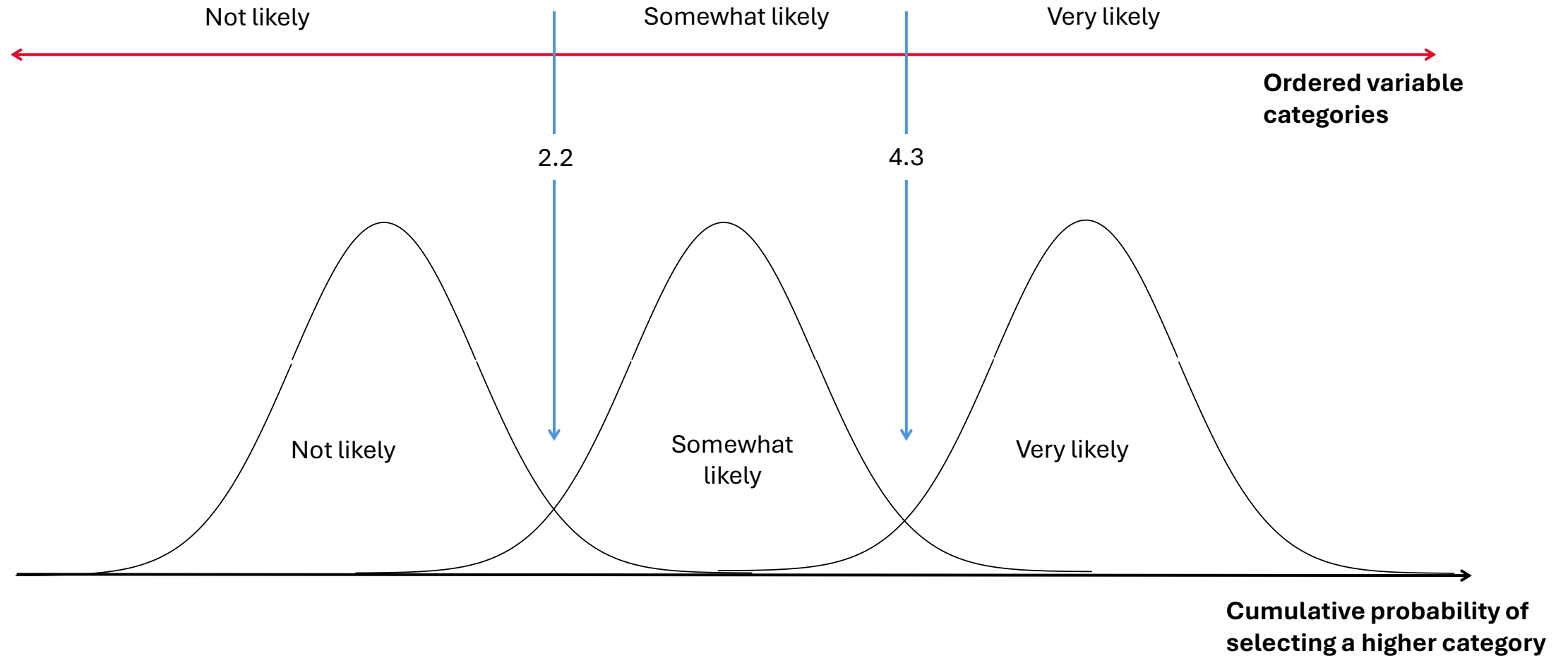
Akaike Information Criterion (AIC).

Penalises model complexity. A lower AIC reflects a better balance between goodness-of-fit and parsimony.

These cut-points determine where, along the linear predictor, **the cumulative probabilities transition from one category to the next.**

Ordinal regression

Estimated Intercepts (Cut-points)



Ordinal regression

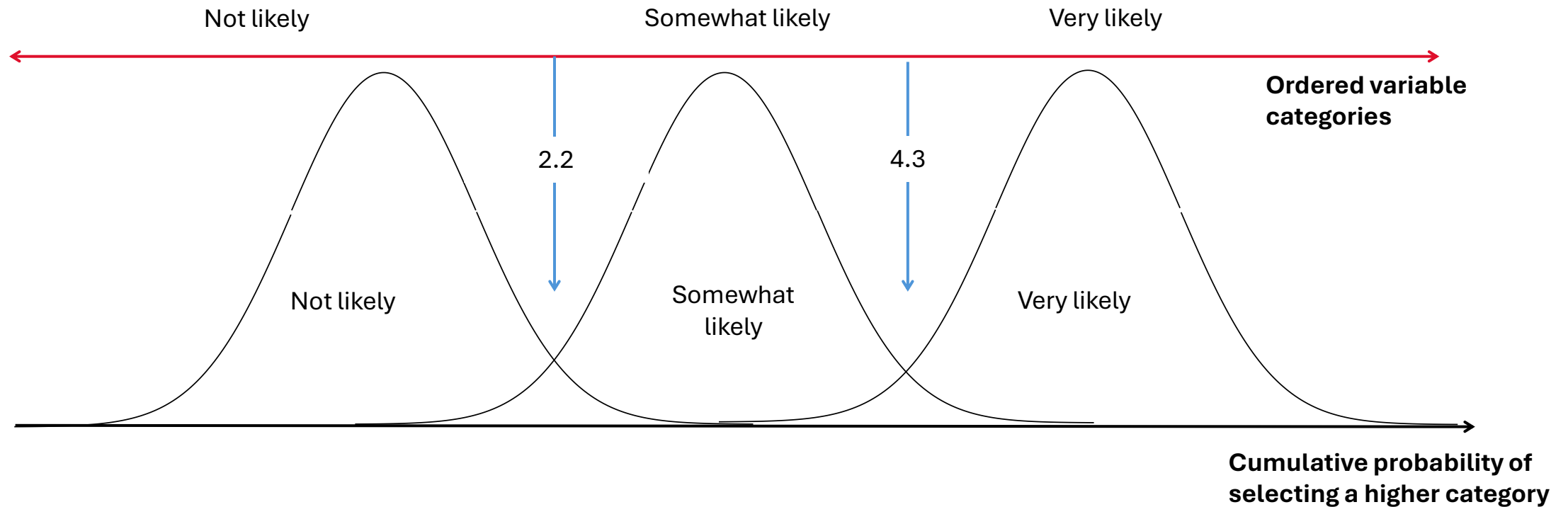
Estimated Intercepts (Cut-points)

The estimated cut-points at 2.2 and 4.3

determine the boundaries separating “not likely” from “somewhat likely”, and “somewhat likely” from “very likely”, respectively.

The categories of the ordinal response variable correspond to intervals on the underlying

latent scale. These latent thresholds do not represent direct probabilities; instead, they indicate the points at **which the cumulative probability of selecting a higher category crosses specific values.**



Ordinal regression

Results of the Ordinal Logistic Regression on the Intention to Apply to a Postgraduate Programme

Variable	Coefficient	Standard Error	t-Statistic	
pared	1.048	0.266	3.942	$p < 0.001^{***}$
public	-0.059	0.298	-0.197	$p = 0.844$
gpa	0.616	0.261	2.363	$p = 0.018^*$
Category Thresholds				
unlikely somewhat likely	2.204	0.780	2.827	$p = 0.005^{**}$
somewhat likely very likely	4.299	0.804	5.345	$p < 0.001^{***}$
Significance codes: *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$				
Residual deviance: 717.02				
AIC: 727.02				

$$\ln \left(\frac{P(Y_n \leq j)}{P(Y_n > j)} \right) = \mu_j - \beta^\top \mathbf{x}_n$$

- **Having parents with higher education.** This factor reduces the cumulative log-odds of being in a lower application-intention category compared with being in the highest category (“very likely”) by 1.048 units. **In substantive terms, students whose parents hold a university degree are more likely to report a higher intention to apply.**
- **Graduating from a public university.** Shows a coefficient that is slightly positive (0.059) but statistically insignificant. Therefore, there is no evidence that attending a public university affects the intention to apply to a postgraduate programme.
- **Grade Point Average (GPA).** Each additional point in GPA decreases the cumulative log-odds of being in a lower category by 0.616 units, indicating that higher academic achievement is associated with higher intention to apply.

The estimated model includes threshold (cut-point) parameters and cumulative probabilities for the lower-order categories. These cut-points indicate the values of the latent variable at which the probability of selecting a higher category exceeds that of the adjacent lower category.

$$\ln \left(\frac{\hat{P}(Y \leq 1)}{\hat{P}(Y_n > 1)} \right) = 2.204 - (1.048 \cdot \text{pared} + -0.059 \cdot \text{public} - 0.616 \cdot \text{gpa})$$

$$\ln \left(\frac{\hat{P}(Y \leq 2)}{\hat{P}(Y_n > 2)} \right) = 4.299 - (1.048 \cdot \text{pared} + -0.059 \cdot \text{public} - 0.616 \cdot \text{gpa})$$

To aid interpretation, the exponentiated coefficients (odds ratios) should be computed. These measure the multiplicative change in the cumulative odds of reporting a higher intention to apply, associated with a one-unit increase in each predictor.

$$e^{\ln \left(\frac{P(Y_n \leq j)}{P(Y_n > j)} \right)} = e^{(\mu_j - \beta^\top \mathbf{x})} = \frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})}$$

$$\frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})}$$

$$\frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})} \longrightarrow e^{(\mu_j - \beta^\top \mathbf{x} + \beta_j)} = e^{(\mu_j - \beta^\top \mathbf{x})} \cdot e^{(\beta_j)}$$

Ordinal regression

Results of the Ordinal Logistic Regression Regarding the Intention to Apply to Postgraduate Studies

Variable	OR	Lower limit	Upper limit
pared	2.8511	1.6958	4.8171
public	0.9429	0.5209	1.6806
gpa	1.8514	1.1136	3.0985

Note: 95% confidence intervals for the odds ratios.

To interpret the results, we may (and should) also apply the exponential function to the estimated parameters to obtain the odds ratios (ORs):

$$e^{\ln\left(\frac{P(Y_n \leq j)}{P(Y_n > j)}\right)} = e^{(\mu_j - \beta^\top \mathbf{x})} = \frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})}$$

$$\frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})}$$

$$\frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})} \longrightarrow e^{(\mu_j - \beta^\top \mathbf{x} + \beta_j)} = e^{(\mu_j - \beta^\top \mathbf{x})} \cdot e^{(\beta_j)}$$

Parental education

For students, whose parents completed higher education, **the probabilities of being in a higher category of intention to apply** (that is, “somewhat likely” or “very likely”, compared with “not likely”) **are 2.85 times greater than for students whose parents do not hold a university degree, holding all other variables constant.**

This corresponds to an increase of approximately 185%:

$$(2.8511 - 1) \times 100 \approx 185\%$$

Using the lower cumulative odds (Logit de $P(Y \leq j)$) the inverse odds ratio is:

$$OR_{\text{Inferior}} = 1/OR$$

$$OR_{\text{Inferior}} = \frac{1}{2.85} \approx 0.35$$

Thus, the lower cumulative odds (“not likely” vs “somewhat or very likely”) are 65% lower for students whose parents have higher education:

$$(1 - 0.35) \times 100 \approx 65\%$$

For students whose parents did attend university, **the probabilities of being in a lower category of intention to apply** are 65% lower than for students whose parents do not have higher education.

Ordinal regression

Results of the Ordinal Logistic Regression Regarding the Intention to Apply to Postgraduate Studies

Variable	OR	Lower limit	Upper limit
pared	2.8511	1.6958	4.8171
public	0.9429	0.5209	1.6806
gpa	1.8514	1.1136	3.0985

Note: 95% confidence intervals for the odds ratios.

To interpret the results, we may (and should) also apply the exponential function to the estimated parameters to obtain the odds ratios (ORs):

$$e^{\ln\left(\frac{P(Y_n \leq j)}{P(Y_n > j)}\right)} = e^{(\mu_j - \beta^\top \mathbf{x})} = \frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})}$$

$$\frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})}$$

$$\frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})} \longrightarrow e^{(\mu_j - \beta^\top \mathbf{x} + \beta_j)} = e^{(\mu_j - \beta^\top \mathbf{x})} \cdot e^{(\beta_j)}$$

Type of University

For students graduating from a public university, **the probabilities of being in a higher intention-to-apply category** (that is, “somewhat likely” or “very likely”, as opposed to “not likely”) are 0.9429 times those of students graduating from private universities, holding all other variables constant.

This corresponds to a 5.71% decrease:

$$(1 - 0.9429) \times 100 \approx 5.71\%$$

Using the lower cumulative odds (Logit de $P(Y \leq j)$) the inverse odds ratio is:

$$OR_{\text{Inferior}} = 1/OR$$

$$OR_{\text{Inferior}} = \frac{1}{0.9429} \approx 1.06$$

Thus, the lower cumulative odds (“not likely” vs. “somewhat or very likely”) are 6% higher for students graduating from public universities.

$$(1.06 - 1) \times 100 \approx 6\%$$

For students who graduated from a public university, **the probabilities of being in a lower category of intention to apply** are 6% higher than those of students who graduated from private universities.

Ordinal regression

Results of the Ordinal Logistic Regression Regarding the Intention to Apply to Postgraduate Studies

Variable	OR	Lower limit	Upper limit
pared	2.8511	1.6958	4.8171
public	0.9429	0.5209	1.6806
gpa	1.8514	1.1136	3.0985

Note: 95% confidence intervals for the odds ratios.

To interpret the results, we may (and should) also apply the exponential function to the estimated parameters to obtain the odds ratios (ORs):

$$e^{\ln\left(\frac{P(Y_n \leq j)}{P(Y_n > j)}\right)} = e^{(\mu_j - \beta^\top \mathbf{x})} = \frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})}$$

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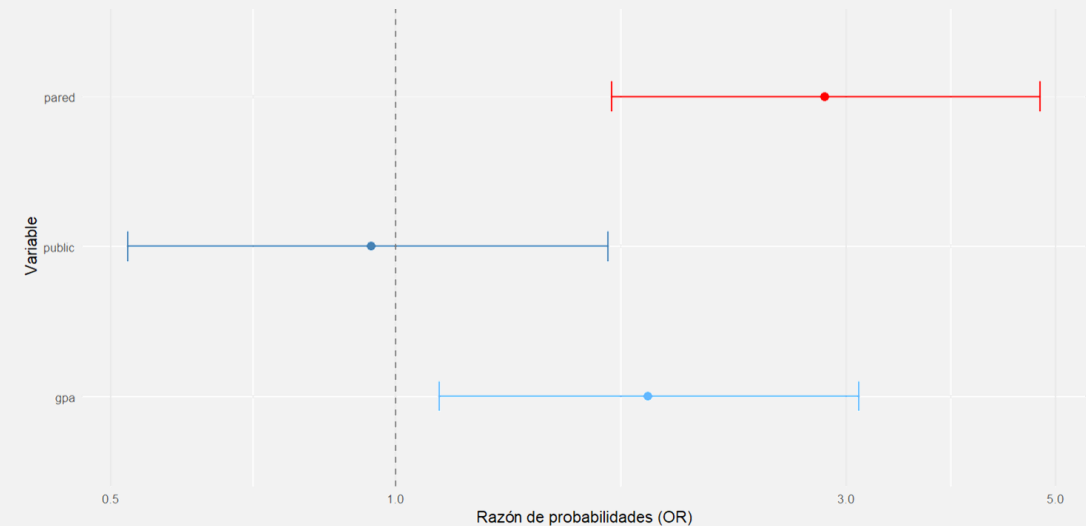
$$\frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \beta^\top \mathbf{x})} \rightarrow e^{(\mu_j - \beta^\top \mathbf{x} + \beta_j)} = e^{(\mu_j - \beta^\top \mathbf{x})} \cdot e^{(\beta_j)}$$

GPA

For each additional unit of GPA, the **probabilities of being in a higher intention-to-apply category** increase by a factor of 1.85, which corresponds to an 85% increase, holding all other variables constant.

$$(1.8514 - 1) \times 100 \approx 85\%$$

Forest Plot of the Estimated Parameters in the Ordinal Model



Ordinal regression

Proportional Odds assumption

To verify whether the effect of any predictor variable (X) on the cumulative odds of obtaining a higher response ($Y > j$) is the same across all threshold points ($\mu_1, \mu_2, \dots$), we estimate auxiliary binary logit models based on dichotomisations of the ordinal dependent variable:

$$1 \quad \frac{p}{1-p} = e^{(\beta^\top \mathbf{x})} \quad Y > 1 = \begin{cases} 1 & \text{if } y = \text{"somewhat likely"} \text{ or "very likely"} \\ 0 & \text{if } y = \text{"not likely"} \end{cases}$$

$$2 \quad \frac{p}{1-p} = e^{(\beta^\top \mathbf{x})} \quad Y > 2 = \begin{cases} 1 & \text{if } y = \text{"very likely"} \\ 0 & \text{if } y = \text{"not likely"} \text{ or "somewhat likely"} \end{cases}$$

Auxiliary logit models

Model	Variable	OR	Lower limit	Upper limit	p-value
$y > 1$	Intercept	0.138	0.028	0.676	0.0146
$y > 1$	pared	2.89	1.61	5.17	< 0.001***
$y > 1$	public	0.818	0.450	1.490	0.5113
$y > 1$	gpa	1.73	1.01	2.95	0.0442*
$y > 2$	Intercept	0.0086	0.0006	0.134	< 0.001***
$y > 2$	pared	2.50	1.18	5.29	0.0167*
$y > 2$	public	1.71	0.749	3.89	0.2034
$y > 2$	gpa	2.09	0.853	5.11	0.1071

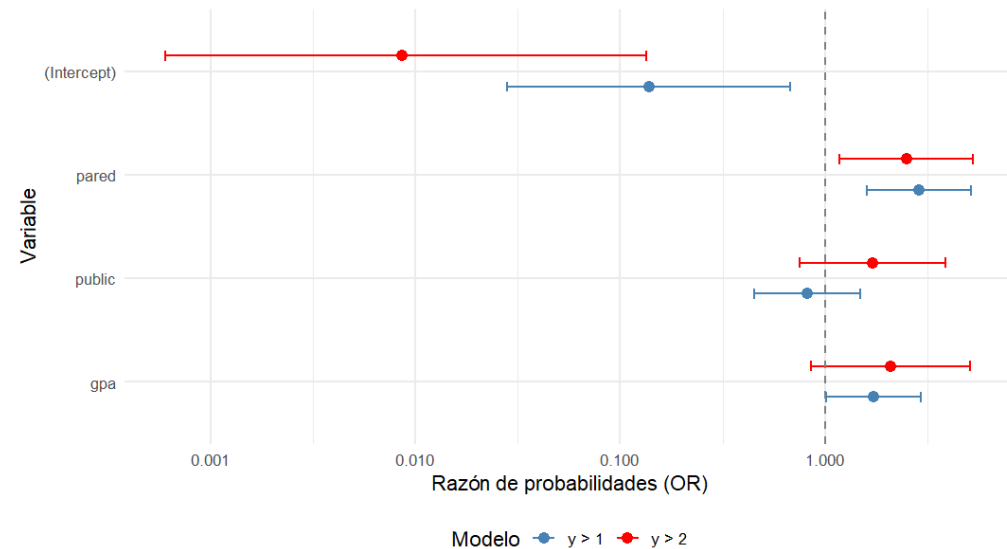
Significance codes: *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$

Note: Each logit model is estimated on a different dichotomisation of the ordinal variable apply.

Model $y > 1$: compares "somewhat likely" and "very likely" against "not likely".

Model $y > 2$: compares "very likely" against "somewhat likely" and "not likely".

Forest Plot of the Estimated Parameters in the Auxiliary Logit Models



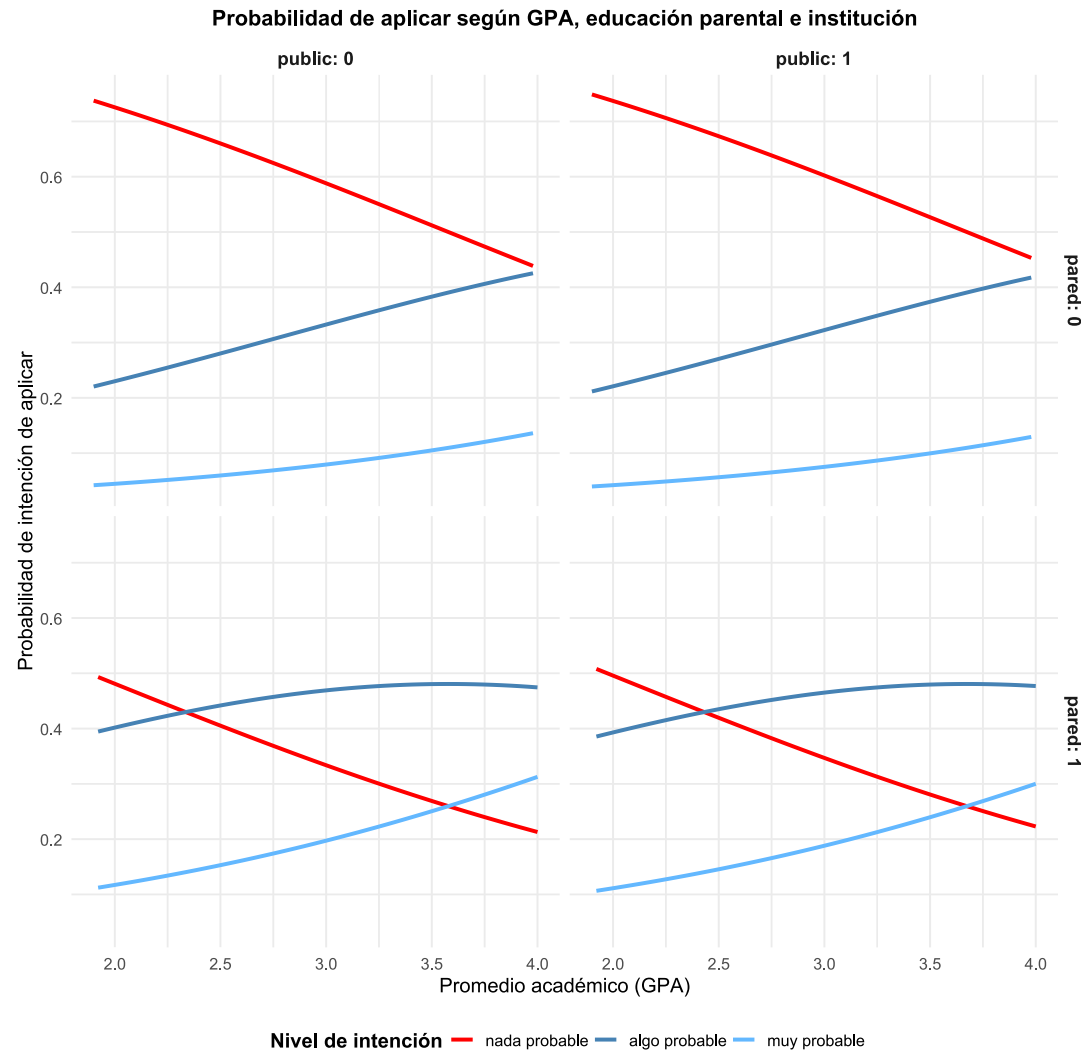
As an additional step, a Brant test may be used to evaluate formally whether the proportional-odds (parallel-regression) assumption is satisfied. The test compares the coefficients estimated in the separate binary logit models for each threshold (e.g., $Y > 1$ and $Y > 2$).

Test	Chi-square (X^2)	df	p-value
Omnibus	4.34	3	0.23
pared	0.13	1	0.72
public	3.44	1	0.06
gpa	0.18	1	0.67

Null hypothesis (H0): The proportional-odds (parallel regression) assumption holds.

Interpretation: High p-values indicate insufficient evidence to reject H0.

Ordinal regression



The logistic model not only enables the estimation of individual probabilities but also allows the construction of comparative intention-to-apply profiles.

By holding GPA constant and systematically varying parental education and institution type, it becomes possible to identify specific probability scenarios for belonging to the highest category of intention to apply.



Sebastián

Type of institution: private
Parents: both hold postgraduate degrees
GPA: 3.5

Predicted probability of belonging to the highest intention-to-apply category: 73%



Felipe

Type of institution: public
Parents: neither parent holds an undergraduate degree
GPA: 3.5

Predicted probability of belonging to the highest intention-to-apply category: 47%

On P-values and statistical significance

Use of the p-value in hypothesis testing

In a conservative approach, until proven otherwise, the null hypothesis is assumed to be valid. **When analyzing a hypothesis test, we want to know:**

What is the probability of obtaining results like those observed in the sample if the null hypothesis were actually true?

This probability is the p-value.

A p-value < 0.05 means that: there is only a 5% or lower probability that a result like the one observed in the sample would occur under the null hypothesis.

If this happens, we should reject the null hypothesis that there is no treatment effect and conclude that, given the available data, the treatment had an effect. **In other words, the treatment effect was significantly different from zero.**

- ✗ If you see a p-value of 0.01, this does not mean that there is a 1% probability that the result is due to chance.
- ✓ The p-value represents the probability of observing the results obtained, or more extreme results, assuming that the null hypothesis is actually true.

Use of the p-value in hypothesis testing

Researchers are interested in whether **exposure to online fake news** is associated with **lower trust in democratic institutions**.



Fake news consumption

Frequency with which individuals are exposed to or engage with online misinformation (e.g. self-reported exposure, interaction with known misinformation sources).



Institutional trust

An index measuring trust in institutions such as parliament, courts, government, or electoral authorities.

H₀ Null hypothesis

Fake news consumption has no association with institutional trust.

H_a Alternative hypothesis

Fake news consumption is associated with institutional trust.

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$$\text{Institutional trust} = \beta_0 + \beta_1 \text{Fake news consumption} + \mathbf{X}_i' \boldsymbol{\gamma} + \varepsilon_i$$

DV: institutional trust	
Fake news consumption	-0.315***
Constant	3.420***
Observations	1,200
Controls	Yes
R ²	0.28

Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.10.

Controls include age, education, income, and political interest.

- The estimated coefficient for fake news consumption is negative.
- The associated p-value is 0.002.

Use of the p-value in hypothesis testing

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- The estimated coefficient for fake news consumption is negative.
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If the null hypothesis were true, this would mean that in repeated random samples drawn from the same population, only 2 out of every 1,000 samples would produce an estimated relationship as large as, or larger than, the one observed in our sample.

Because this probability is very small, we reject the null hypothesis and conclude that higher fake news consumption is significantly associated with lower institutional trust.

Use of the p-value in hypothesis testing

Why is the p-value problematic and how can it lead to Type I errors?

We generally use $\alpha = 0.05$ as the criterion for statistical significance, **but this value is arbitrary**, and it is important to understand how it came to be used.

Fisher made numerous fundamental contributions to statistics and **developed the concept of significance testing**, in which the p-value is used as a guide to reach a conclusion about a hypothesis.

- ✓ Fisher used the term “hypothesis” to refer specifically to the null hypothesis.
- ✓ If $p < 0.01$, he generally regarded the result as clearly significant.
- ✓ He used 0.05 as a reference point, but for a broader range of p-values, possibly between 0.01 and 0.20, he often recommended additional experiments to further investigate the observed effect.

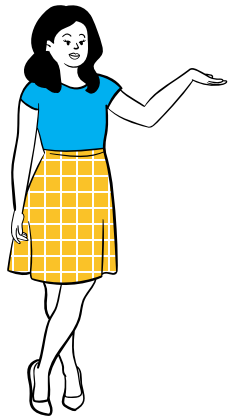


Ronald Fisher
(1890-1962)

Fisher viewed the p-value as a measure of the strength of evidence against the hypothesis:
the smaller the p-value, the stronger the reason to doubt the null hypothesis

How can we evaluate whether an observed result provides genuine evidence against a hypothesis, rather than being a coincidence produced by random variation?

Why is the p-value problematic and how can it lead to Type I errors?



A person claims to be able to distinguish, purely by taste, **whether a cup of tea was prepared by adding milk first and then tea, or adding tea first and then milk.**



Does the person genuinely possess this ability?
Are correct answers simply the result of guessing?

Rather than relying on intuition or authority, **Fisher proposes that the claim should be evaluated using a formal experimental procedure.**

He did not attempt to prove directly that the person has the ability. Instead, he reframed the problem like :

How likely would this level of performance be if the person had no real ability at all?

This question lies at the heart of the p-value concept.

Why is the p-value problematic and how can it lead to Type I errors?

How likely *would this level of performance* be if the person *had no real ability at all*?

H₀ Null hypothesis

The person has no ability to distinguish the order in which tea and milk were poured. Any correct classification occurs purely by chance.

- ✓ There is no association between how a cup was prepared and how it is identified.
- ✓ The person's responses are equivalent to random guessing.

The experiment structure



Four cups with milk **poured first (MF)**

Four cups with tea poured first **(TF)**

Why is the p-value problematic and how can it lead to Type I errors?

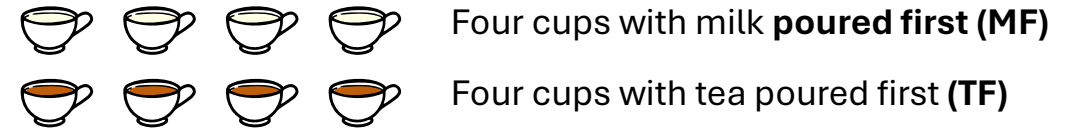
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The experiment structure



Then all 8 cups were ordered randomly



The participant is asked to **classify each cup according to how it was prepared**. Importantly, the participant **knows** that there are exactly four cups of each type

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The experiment structure

The perfect classification of the tea would look like this:

		Guess		
				
Reality		4	0	4
		0	4	4
		4	4	8

Possible outcomes include:

- Four correct classifications (what we would expect on average under random guessing)
- Five or six correct classifications (somewhat better than chance)
- Seven or eight correct classifications (highly unlikely under chance alone)

Why is the p-value problematic and how can it lead to Type I errors?

The experiment structure

Success number (k)	Permutations of selection	Number of combinations
0	oooo	$1 \times 1 = 1$
1	ooox, ooxo, oxoo, xooo	$4 \times 4 = 16$
2	ooxx, oxox, oxxo, xoxo, xxoo, xoox	$6 \times 6 = 36$
3	oxxx, xoxx, xxox, xxxo	$4 \times 4 = 16$
4	xxxx	$1 \times 1 = 1$
		70

- Correct guess (x)
- Incorrect guesses (o)
- The correct guess can appear in any position
- There are 4 items of one type and 4 of the other type
- Under the null hypothesis, true assignments and guesses are independent

For a given number of successes k :

Number of combinations = $\binom{4}{k} \binom{4}{4-k}$

Ways to choose which **true items** are of one type.
which cups are actually milk-first

Ways to choose which items **are guessed** as that type.
which cups are guessed to be milk-first.

If k guesses align with the true milk-first cups, the remaining $4-k$ guesses must be wrong

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- Correct guess (x)
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Let's see what happens when $K=1$ success

Number of combinations = $\binom{4}{1} \binom{4}{4-1}$

- Choose which 1 of the 4 true milk-first cups will be correctly identified
- Choose which 3 of the 4 tea-first cups will be incorrectly guessed as milk-first

Number of combinations = $\binom{4}{1} \binom{4}{4-1} = 4 \times 4 = 16$

Once we have:

The total number of permutations

The Number of permutations at least as extreme as the observed result

We can compute:

$$p - value = \frac{\text{extreme permutations}}{\text{all permutations}}$$

Why is the p-value problematic and how can it lead to Type I errors?



Jerzy Neyman
(1894-1981)

Neyman and Pearson suggested that a p-value < 0.05 implies **that the data provide sufficient evidence to reject the null hypothesis and conclude that an effect is statistically significant.**

⚠ However, the use of the Neyman–Pearson significance criterion with $\alpha = 0.05$ can be problematic **and carries a high probability of producing Type I errors, unless certain conditions are met when designing the study and analysing the data:**

- ✓ The sample size must be determined in advance.
- ✓ The hypotheses and the analysis used to test them must be defined beforehand.
- ✓ Potential control variables must be identified prior to the analysis.
- ✓ Criteria for excluding participants and handling missing data must be specified in advance.
- ✓ For exploratory analyses (that is, analyses that are not hypothesis-testing), a more stringent significance criterion should be used.

What happens when the sample size is not determined in advance?

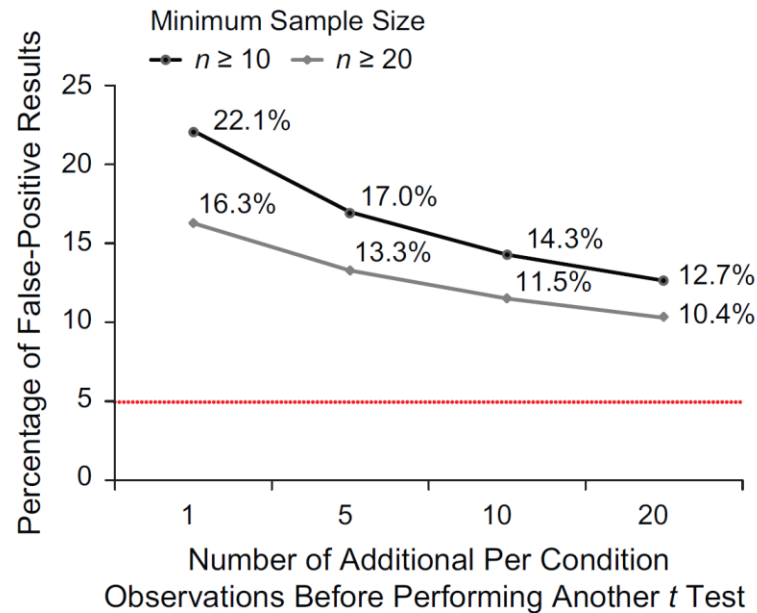


Fig. 1. Likelihood of obtaining a false-positive result when data collection ends upon obtaining significance ($p \leq .05$, highlighted by the dotted line). The figure depicts likelihoods for two minimum sample sizes, as a function of the frequency with which significance tests are performed.

According to Simmons et al. (2011), failing to decide the sample size (or the rule for stopping data collection) before starting the study is an example of “**researcher degrees of freedom**”.

This practice, which involves collecting data based on preliminary analyses, **can dramatically increase the actual false-positive rate** beyond the nominal 5% threshold ($p \leq 0.05$).

For example, a researcher who begins with 10 observations per condition and tests for significance after adding each new observation finds a significant effect in 22% of cases.

This demonstrates **that flexibility in choosing the sample size, especially stopping data collection once significance is reached, can unacceptably facilitate the publication of “statistically significant” evidence for a false hypothesis.**

Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant. *Psychological Science*, 22(11), 1359–1366. <https://doi.org/10.1177/0956797611417632>

P-hacking

The table on the right is based on a computational simulation of an experiment that investigated the impact of an independent variable X, which consists of two conditions (20 participants per condition), on two dependent variables (Y and Z).

- **Situation A:** shows the probability of obtaining false positives (for different significance levels) when researchers test the effect of X on both dependent variables, but report the result for only one analysis.
- **Situation B:** indicates the probability of obtaining false positives if researchers add 10 additional observations per condition after initially non-significant results.
- **Situation C:** reflects the probability of obtaining false positives when researchers include a control variable, such as gender, in the analysis.
- **Situation D:** shows the probability of obtaining false positives if researchers divide the independent variable into three conditions (for example, low, medium, and high) and then report results for only two of them.

Researcher degrees of freedom	Significance level		
	$p < .1$	$p < .05$	$p < .01$
Situation A: two dependent variables ($r = .50$)	17.8%	9.5%	2.2%
Situation B: addition of 10 more observations per cell	14.5%	7.7%	1.6%
Situation C: controlling for gender or interaction of gender with treatment	21.6%	11.7%	2.7%
Situation D: dropping (or not dropping) one of three conditions	23.2%	12.6%	2.8%
Combine Situations A and B	26.0%	14.4%	3.3%
Combine Situations A, B, and C	50.9%	30.9%	8.4%
Combine Situations A, B, C, and D	81.5%	60.7%	21.5%

Note: The table reports the percentage of 15,000 simulated samples in which at least one of a set of analyses was significant. Observations were drawn independently from a normal distribution. Baseline is a two-condition design with 20 observations per cell. Results for Situation A were obtained by conducting three t tests, one on each of two dependent variables and a third on the average of these two variables. Results for Situation B were obtained by conducting one t test after collecting 20 observations per cell and another after collecting an additional 10 observations per cell. Results for Situation C were obtained by conducting a t test, an analysis of covariance with a gender main effect, and an analysis of covariance with a gender interaction (each observation was assigned a 50% probability of being female). We report a significant effect if the effect of condition was significant in any of these analyses or if the Gender $\times$ Condition interaction was significant. Results for Situation D were obtained by conducting t tests for each of the three possible pairings of conditions and an ordinary least squares regression for the linear trend of all three conditions (coding: low = -1, medium = 0, high = 1).

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P-hacking

P-hacking is defined as the selection of statistically significant estimates for publication within a study.

P-hacking occurs when **researchers use flexibility in data collection and analysis to find and report results that meet the conventional threshold for statistical significance ($p \leq 0.05$)**.

Common p-hacking practices

- ✗ **Modifying the set of control variables** can influence the estimation of the effect of interest, allowing researchers to select specifications that generate statistically significant results. This practice is also known as multiple model specification, data exploration, or data mining.
- ✗ **Omitting relevant variables (omitted-variable bias)**. This type of p-hacking can generate statistical patterns that resemble real effects, making it difficult to distinguish between a genuine effect and a false positive caused by b

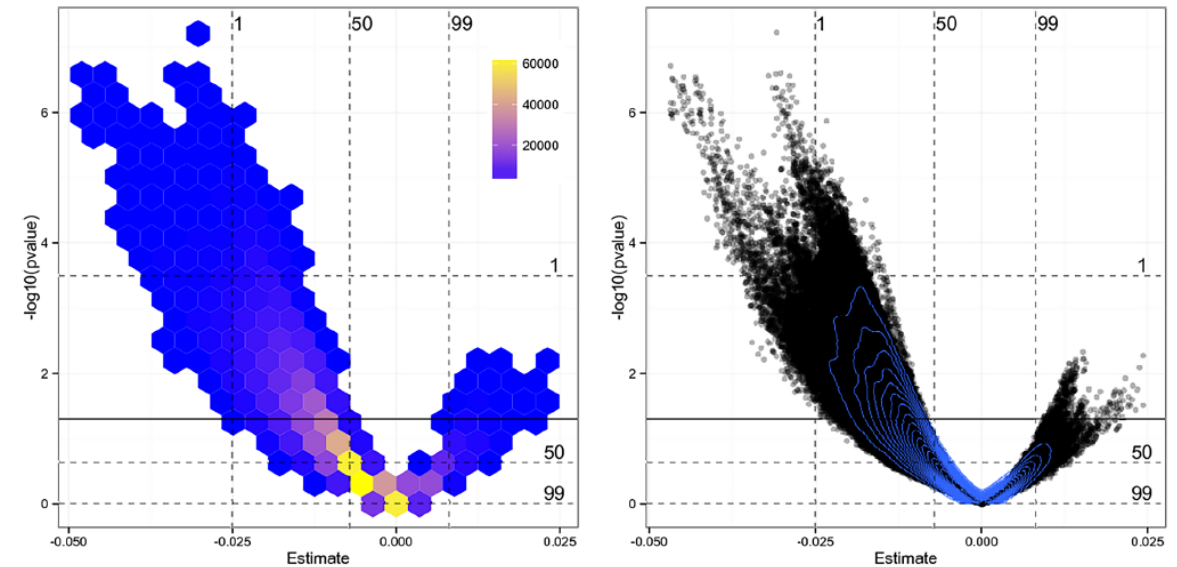


Fig 2. Vibration plot for the effect of malaria prevalence on economic growth. The vibration plot shows estimates of the effect of malaria prevalence in 1966 on the annualized average growth rate of real GDP per capita (1960–1996) on the x-axis. The y-axis shows transformed p -values of these estimates. The plot is based on 100 random samples of countries drawn from a uniform distribution with sample size between 50 and 99. For each sample of countries all 5,005 regression models are estimated resulting in 500,500 estimates of β . The dashed lines represent the 1, 50, and 99 quantiles of the distribution of transformed p -values and of the distribution of β , respectively. The solid line represents $p = 0.05$. Note that due to the transformation of p -values estimates above the line are statistically significant and below the line estimates are insignificant.

Bruns, S. B., & Ioannidis, J. P. A. (2016). P-Curve and p-Hacking in Observational Research. *PLoS ONE*, 11(2), e0149144. <https://doi.org/10.1371/journal.pone.0149144>